

Low-overhead ML methods for beamforming/combining in RIS assisted millimeter-wave massive MIMO systems

Oumaïma Bounhar

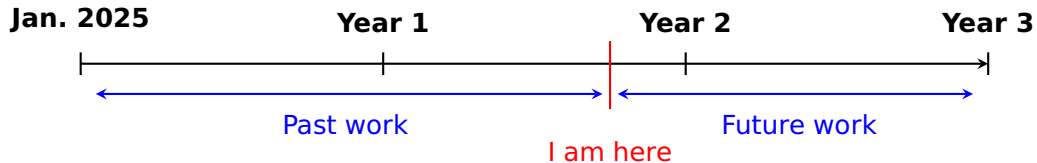
Supervised by Pr. **Ghaya Rekaya-Ben Othman**

Télécom Paris, Institut-Mines-Télécom, Palaiseau, France



22 septembre 2026

Thesis Advancement & Context





Outline

Context

System Model

State-of-the-art

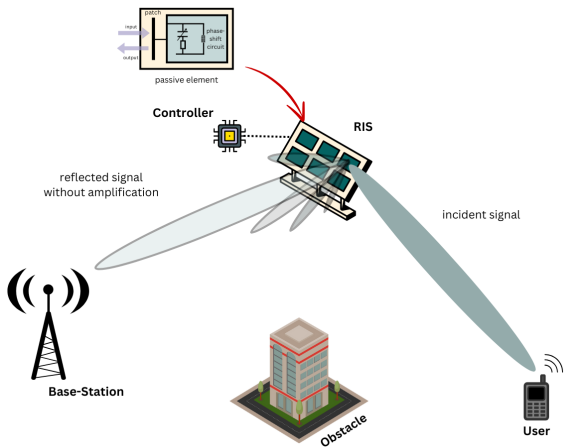
Proposed approach : Deep Reinforcement Learning

Sionna : Realistic channel generation and validation

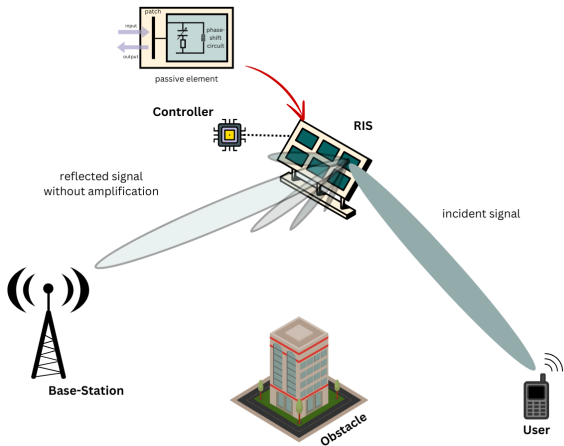
Thesis Vision Board (3 Years)

Future Perspectives

Context

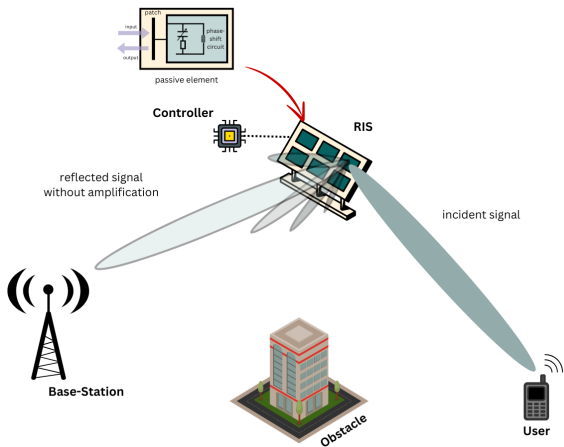


Context



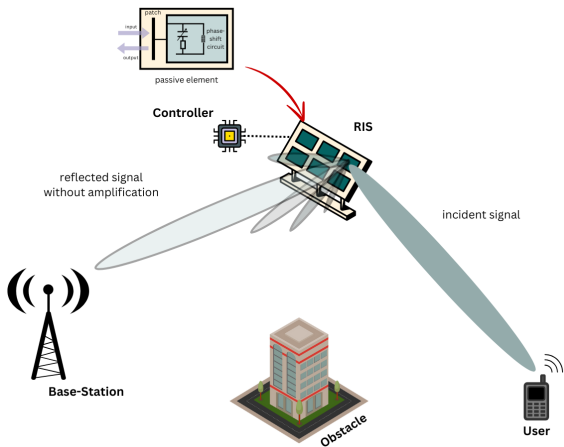
- mmWaves suffer from fading environments;

Context



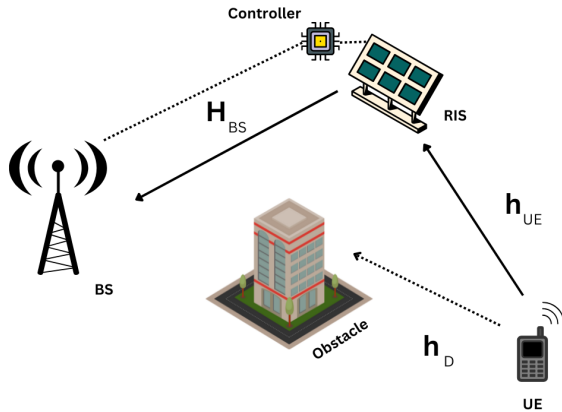
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- 5G & 6G rely on millimeter waves (mmWaves) for higher coverage ;

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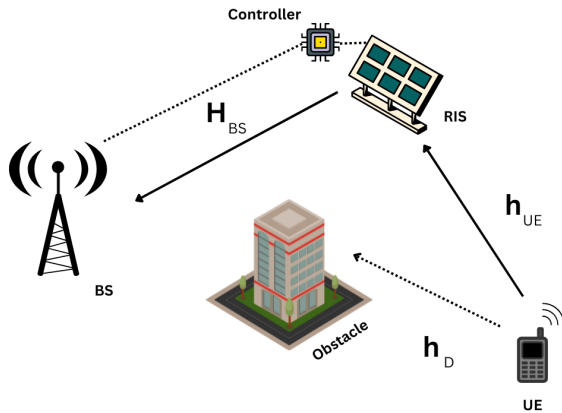
- mmWaves suffer from fading environments ;
- 5G & 6G rely on millimeter waves (mmWaves) for higher coverage ;
- RIS is a promising solution with low power consumption.

System Model



■ $N_{UE} = 1;$

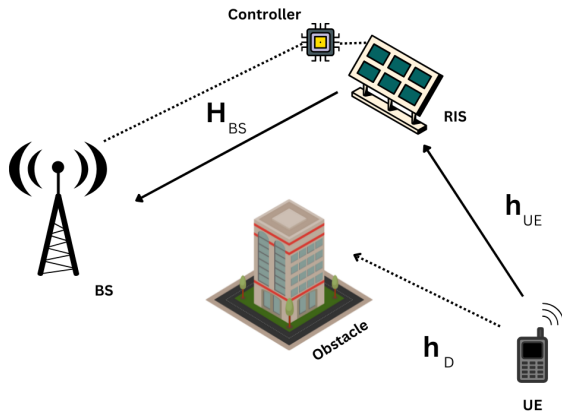
System Model



■ $N_{UE} = 1;$

■ $N_{BS} = 64;$

System Model

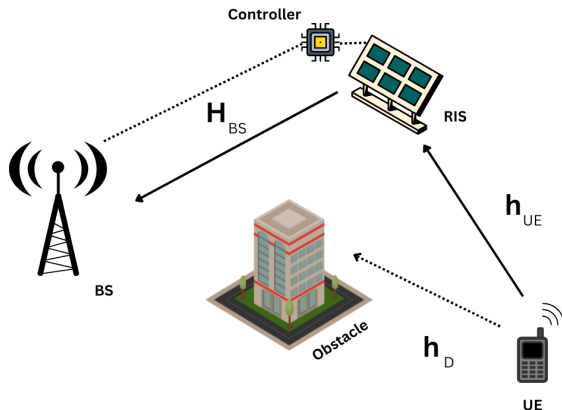


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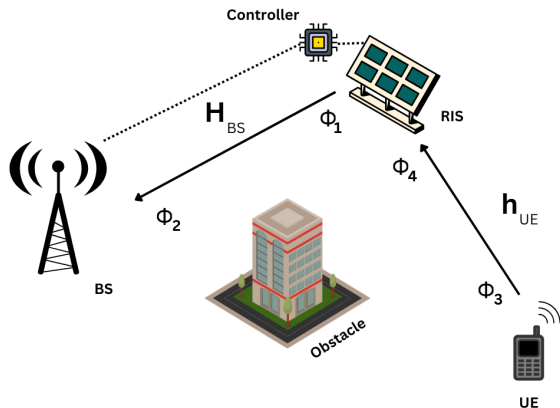
■ $N_{RIS} = 100;$

System Model



- $N_{UE} = 1$;
- $N_{BS} = 64$;
- $N_{RIS} = 100$;
- The direct link is blocked.

System Model



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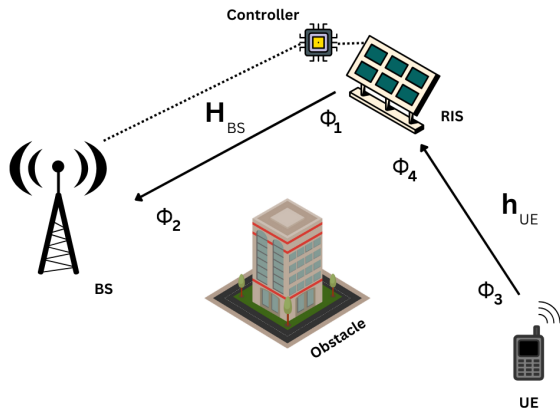
- The direct link is blocked.

We consider half-wave spaced ULAs :

$$\mathbf{H}_{BS} = \mathbf{a}_N(\phi_1)\mathbf{a}_M^H(\phi_2)$$

$$\mathbf{h}_{UE} = \mathbf{a}_1(\phi_3)\mathbf{a}_N^H(\phi_4)$$

System Model



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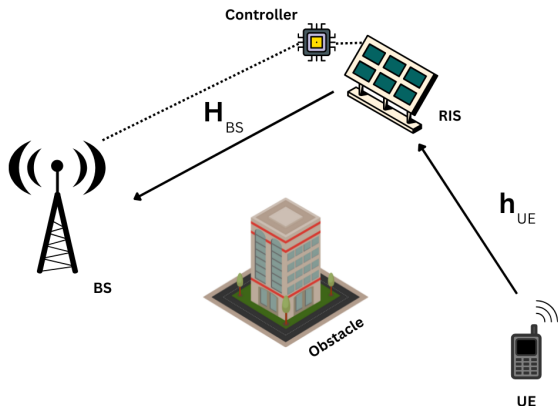
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$$\mathbf{h}_{UE} = \mathbf{a}_1(\phi_3) \mathbf{a}_N^H(\phi_4)$$

where the steering vector is :

$$\mathbf{a}_N(\Phi) = [1, e^{j\pi \sin(\Phi)}, \dots, e^{j\pi(N-1) \sin(\Phi)}]^T$$

System Model

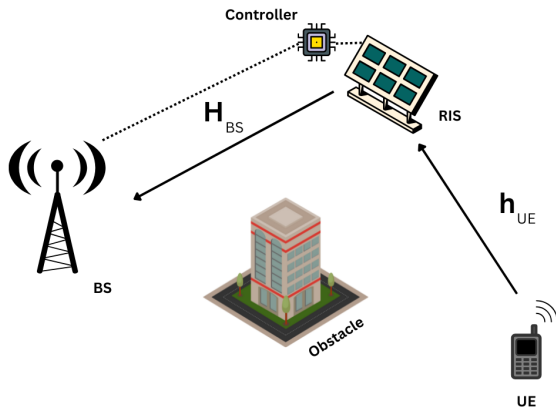


The received signal at the base station (BS) is :

$$\mathbf{y} = \sqrt{P}\mathbf{H}_{\text{BS}}\mathbf{\Theta}\mathbf{h}_{\text{UE}}s + \mathbf{w}$$

where P is the transmit power of the user, $\|s\|^2 = 1$, $\mathbf{w} \sim \mathcal{CN}(0, \sigma_w^2)$ and $\mathbf{\Theta} = \text{diag}(\mathbf{\Phi})$, with $\mathbf{\Phi} = [e^{j\theta_0}, \dots, e^{j\theta_{N_{\text{RIS}}-1}}]$, where the θ_i are the phase shifts introduced by the RIS, $i \in [0, N_{\text{RIS}} - 1]$.

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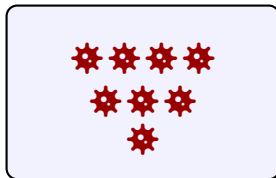
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We write the full channel :

$$\mathbf{h}(\mathbf{\Phi}) = \mathbf{H}_{\text{BS}}\text{diag}(\mathbf{\Phi})\mathbf{h}_{\text{UE}}$$

A Motivating Analogy

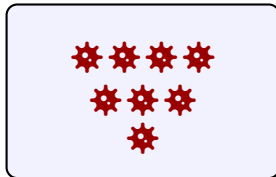
Reference Book (Dataset)



8 known diseases

A Motivating Analogy

Reference Book (Dataset)



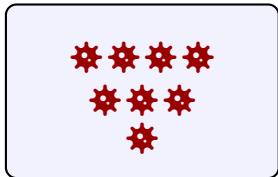
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Clinician

A Motivating Analogy

Reference Book (Dataset)



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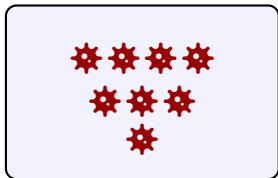
14 Cheap Lab Tests



Clinician

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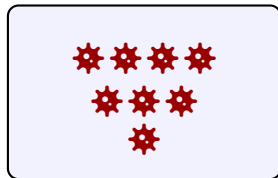
sequentially run, one at a time



Clinician

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compare results

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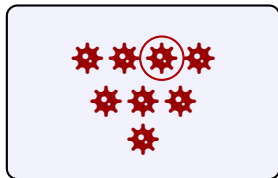
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Clinician

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Reference Book (Dataset)



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Diagnosis found after
only a few tests!

compare results

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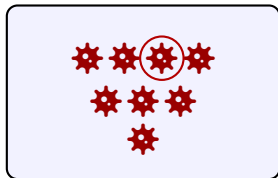
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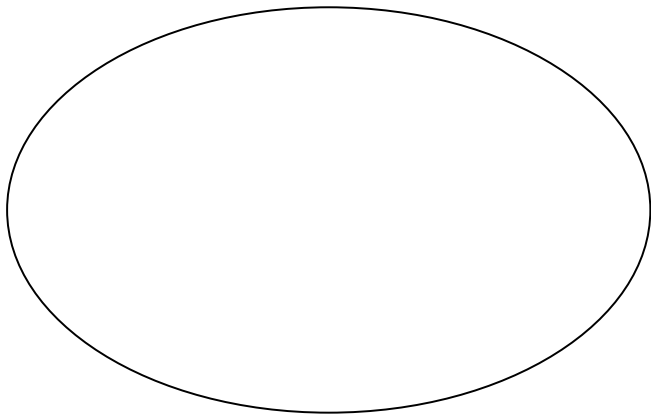
Clinician

Goal : find the right diagnosis using as few cheap tests as possible.

Building the Dataset

How is a sample constructed ?

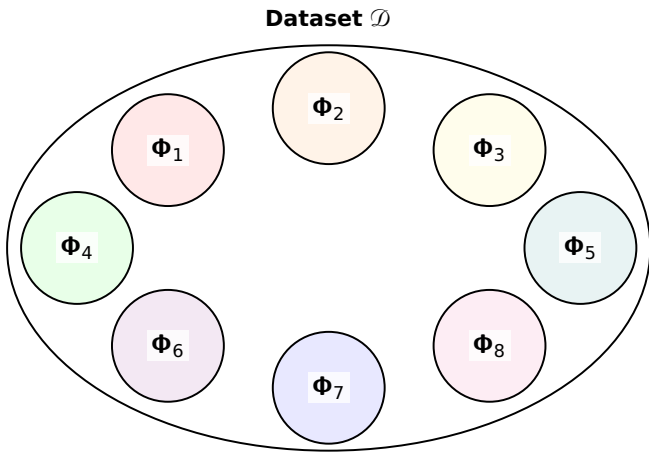
Dataset $\mathcal{D}$



1. Draw a random channel $\mathbf{H}$.

Building the Dataset

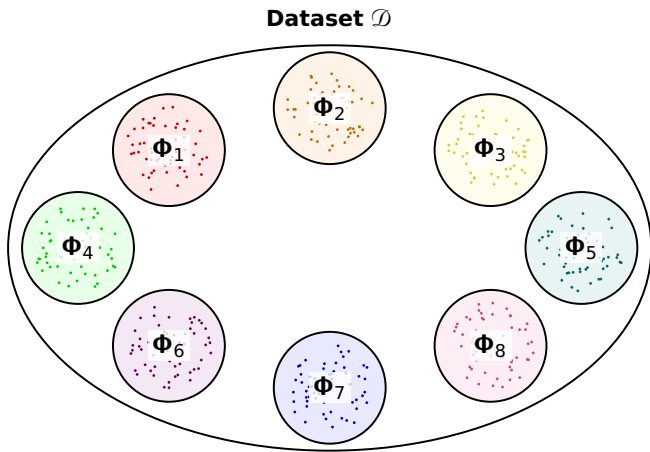
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2. Run all $N_c=8$ communication codewords on $\mathbf{H}$ **without noise** (forbidden tests) $\rightarrow$ best RSE $\rightarrow$ assign c^* .

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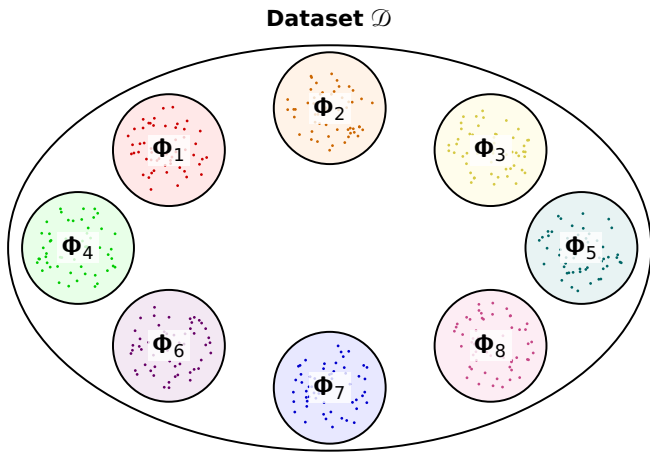
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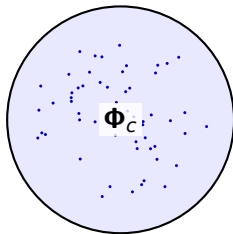


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Each class $\Phi_c \geq 200$ such samples.

The Likelihood Model

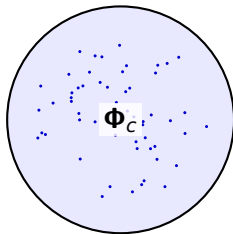
This collection **is** the likelihood model.



The Likelihood Model

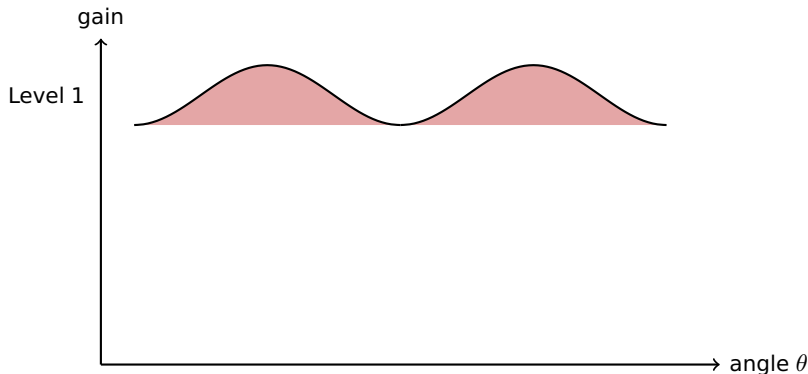
This collection **is** the likelihood model.

“What do the 14 pilot readings
tend to look like
when the answer is c ?”



What Does Each Class Represent?

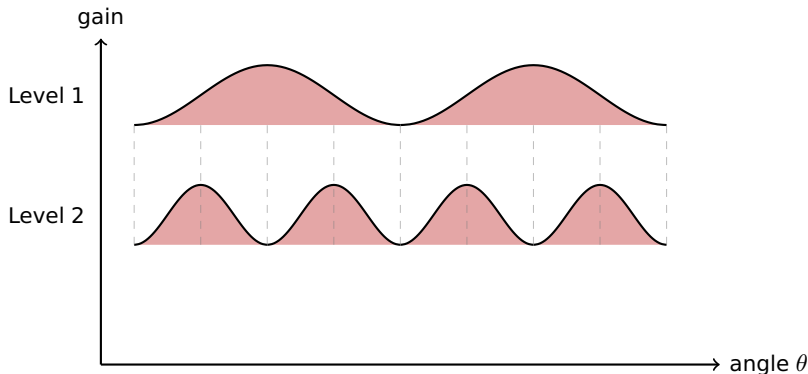
Hierarchical codebook : recursive splitting down to $N_c = 8$ classes



Red : successive pilot search levels, each splitting the angular range in two.

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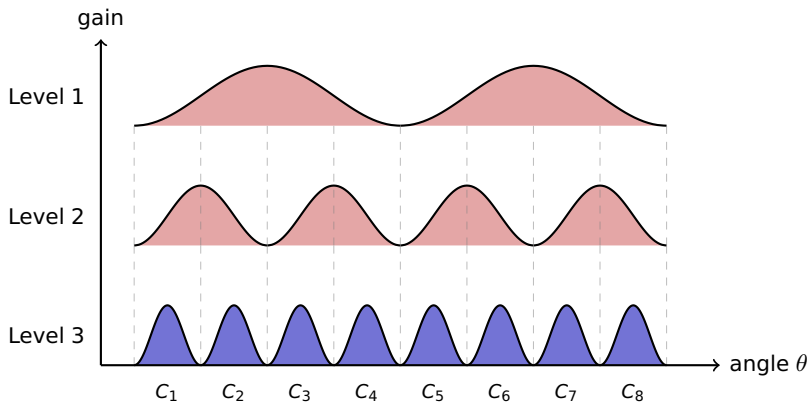
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What Does Each Class Represent ?

Hierarchical codebook : recursive splitting down to $N_c = 8$ classes



Red : successive pilot search levels, each splitting the angular range in two.

Blue : final $N_c = 8$ narrow DFT communication beams Φ_c .

Definition (Rate)

We want to get a higher coverage, which leads to maximizing the rate :

$$R = \max_{i \in \llbracket N_c \rrbracket} \left[\log_2 \left(1 + \frac{P \|\mathbf{H}_{BS} \text{diag}(\boldsymbol{\Phi}_i) \mathbf{h}_{UE}\|_2^2}{\sigma_w^2} \right) \right]$$

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CSI

Channel estimation is challenging.

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CSI

Channel estimation is challenging.

No CSI

"Blind" method.



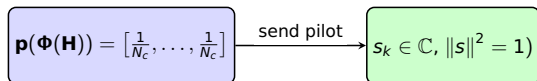
State-of-the-art

Bayesian Classifier : Protocol

$$\mathbf{p}(\Phi(\mathbf{H})) = \left[\frac{1}{N_c}, \dots, \frac{1}{N_c} \right]$$

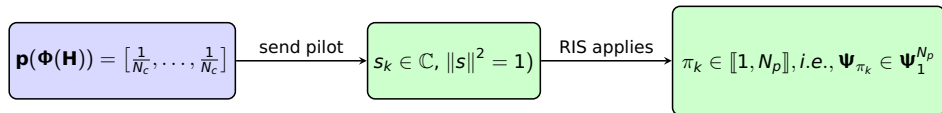
State-of-the-art

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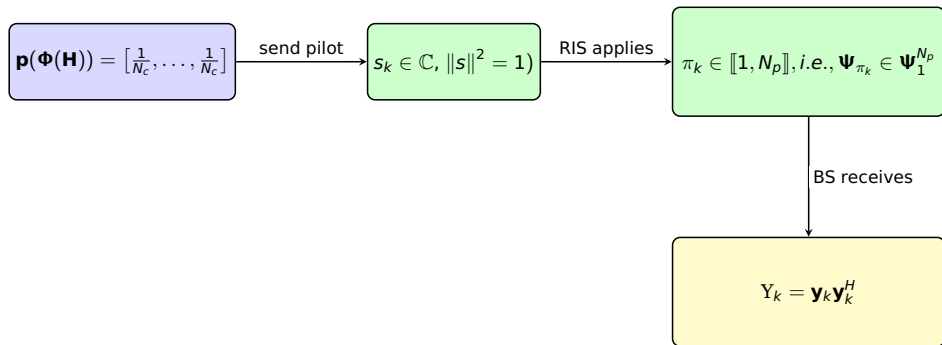
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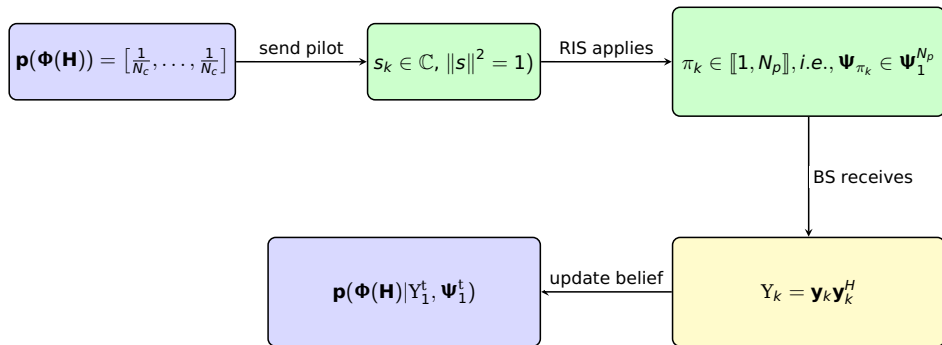
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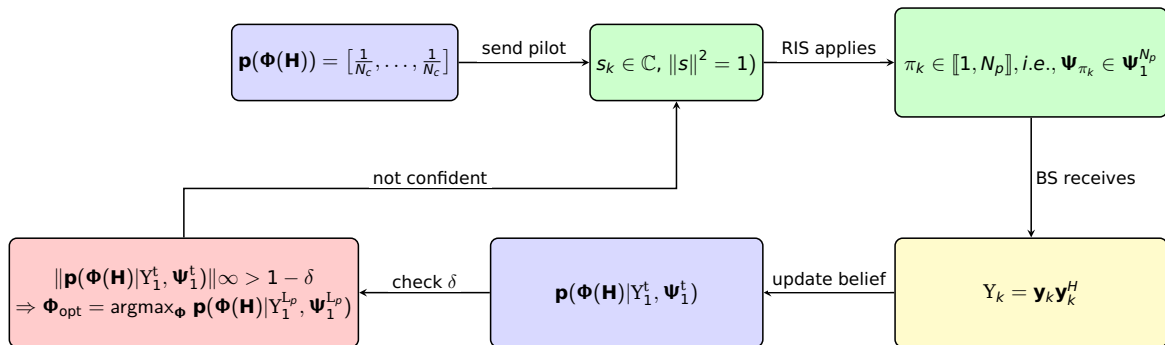
State-of-the-art

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State-of-the-art

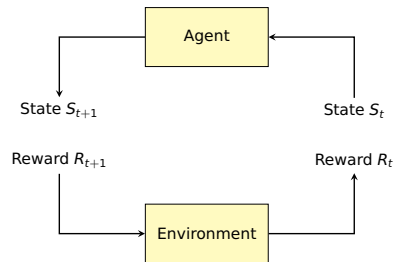
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State-of-the-art

Reinforcement Learning : State, Action, and Reward Flow

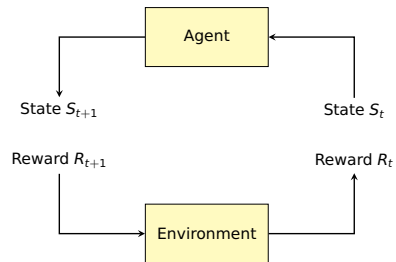
■ Initial state $\mathbf{S}_{\text{init}} \Rightarrow \mathbf{p}(\Phi(\mathbf{H})) = \left[\frac{1}{N_c}, \dots, \frac{1}{N_c} \right]$



State-of-the-art

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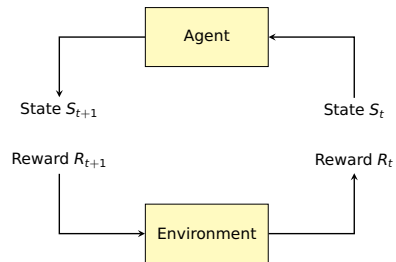
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- Intermediate State $\mathbf{S}_t \rightarrow \mathbf{p}(\Phi(\mathbf{H}) | \mathbf{Y}_1^t, \Psi_1^t)$



State-of-the-art

Reinforcement Learning : State, Action, and Reward Flow

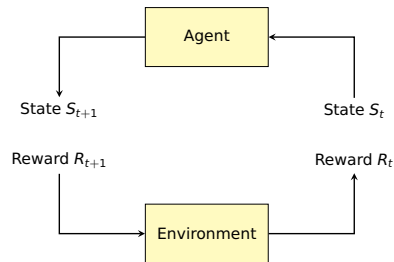
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State-of-the-art

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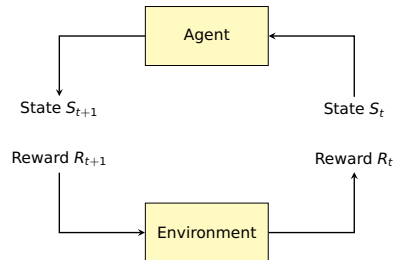
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- Reward $\mathbf{R}_t \rightarrow -1$ until reaching $\mathbf{S}_{\text{terminal}}$ where confidence level is $1 - \delta$



State-of-the-art

Reinforcement Learning : State, Action, and Reward Flow

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Objective

$$\Phi_{\text{opt}} = \operatorname{argmax}_{c \in \llbracket 1, N_c \rrbracket} \mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})$$

$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} > 1 - \delta\}$$

State-of-the-art

Stochastic shortest path problem : Policy, Objective & Q-Learning

Policy

A policy $\pi(S)$ determines which action A is chosen in a given state S :

$$\pi(S) = A$$

State-of-the-art

Stochastic shortest path problem : Policy, Objective & Q-Learning

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Rewritten objective :

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State-of-the-art

Stochastic shortest path problem : Policy, Objective & Q-Learning

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Q-Learning

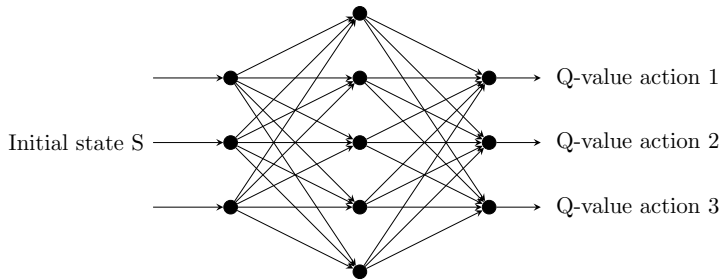
Learn Q -function $Q(S, A)$ from experienced transitions (S, A, R, S') , then act greedily :

$$\underbrace{Q(S, A) \leftarrow Q(S, A) + \alpha [R + \gamma \max_{A'} Q(S', A') - Q(S, A)]}_{\text{update}} \quad \underbrace{\pi^*(S) = \arg \max_A Q^*(S, A)}_{\text{greedy policy}}$$

Proposed approach : Deep Reinforcement Learning

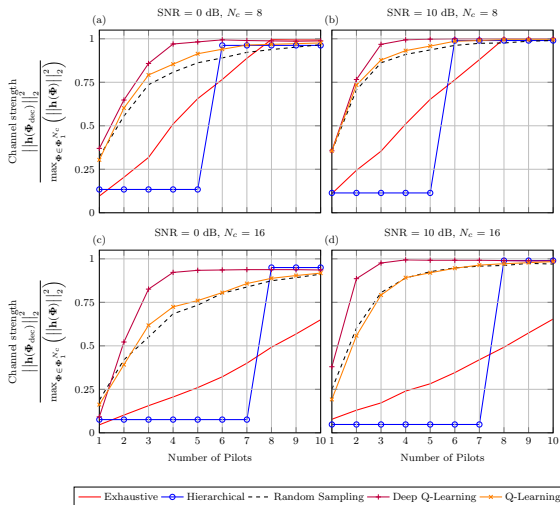
Deep Q-Neural Network

- Replacing the Q-table with a neural network (NN) that generalizes over continuous state spaces.



Proposed approach : Deep Reinforcement Learning

Simulation results : Comparison to other benchmarks



[2] : O. Bounhar, T. Chene, G. R.-B. Othman, and O. Damen, "Deep q-learning for passive beamforming in ris-assisted mimo systems," 2026.



Proposed approach : Deep Reinforcement Learning

Limitations of Deep Q-Learning

- Good for scaling size of codebook and SNR compared to QL
- Couldn't scale for multipath -> move DDQN and PPO, they behaved better than DQN for single path but still not generalizing to multipath
- cite formula of SV multipath (add details to appendix)



Proposed approach : Deep Reinforcement Learning

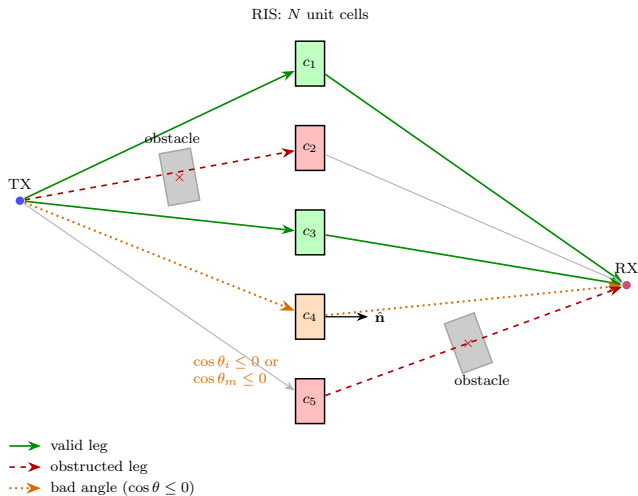
Double Deep Q-Neural Network



Proposed approach : Deep Reinforcement Learning Proximal Policy Optimization

Sionna - Realistic channel generation and validation

Ray tracing : one candidate path per unit cell



Every RIS unit cell c_n , $n = 1, \dots, N$, is treated as an **independent** candidate path :

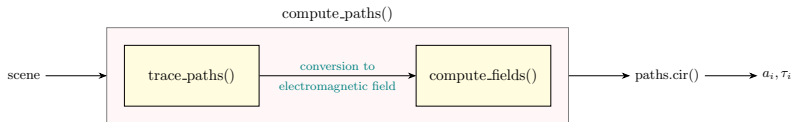
$$\text{TX} \rightarrow c_n \rightarrow \text{RX}.$$

Using the angles from the previous slide, a candidate survives only if :

- both legs are unobstructed ;
- $\cos \theta_i > 0$ and $\cos \theta_m > 0$.

Sionna - Realistic channel generation and validation

From scene geometry to channel taps



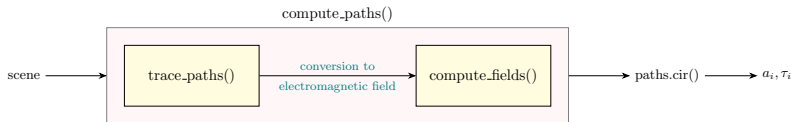
The channel impulse response is reconstructed from the N_{valid} **geometric and deterministic** per-cell taps, either kept individually resolved or coherently combined into a single tap at the mean delay :

Individually resolved (multipath)

$$h_{\text{RIS}}(t) = \sum_{n=1}^{N_{\text{valid}}} \text{coef}_n \delta(t - \tau_n)$$

Sionna - Realistic channel generation and validation

From scene geometry to channel taps



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Coherently combined (single tap)

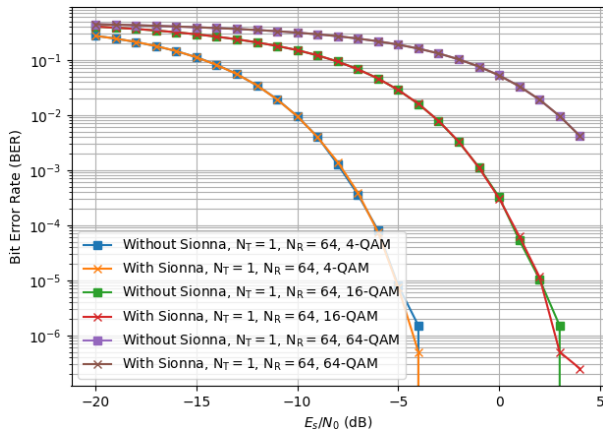
$$\bar{\tau} = \frac{1}{N_{\text{valid}}} \sum_n \tau_n$$

$$h_{\text{RIS}}(t) = a_{\text{RIS}} \delta(t - \bar{\tau})$$

$$a_{\text{RIS}} = \sum_{n=1}^{N_{\text{valid}}} \text{coef}_n e^{-j2\pi f(\tau_n - \bar{\tau})}$$

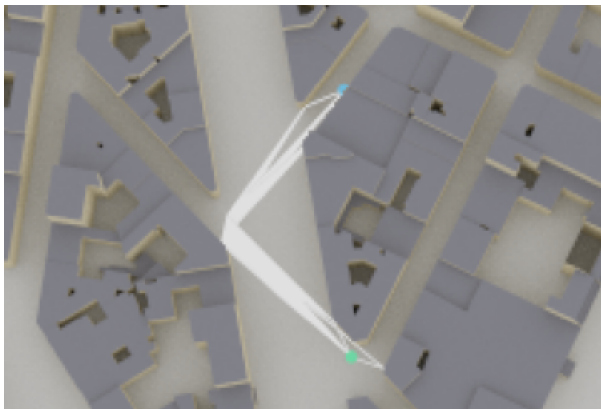
Sionna - Realistic channel generation and validation

BER validation : Sionna vs. theoretical Rayleigh



Sionna - Realistic channel generation and validation

Single-bounce per cell : RIS multipath

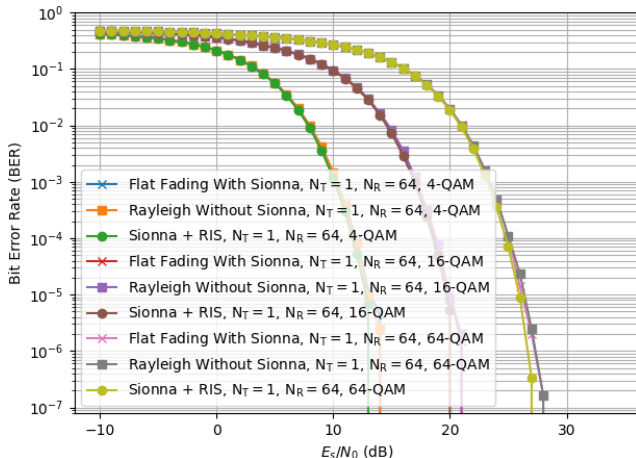


Each traced path is a single-bounce candidate $TX \rightarrow c_n \rightarrow RX$ through one RIS unit cell c_n , obtained by ray tracing (Sionna RT) over the scene. A path is kept only if it satisfies :

- **line-of-sight** on both legs ($TX-c_n$ and c_n-RX unobstructed) ;
- **positive illumination/re-radiation** : $\cos \theta_i > 0$ and $\cos \theta_m > 0$.

Sionna - Realistic channel generation and validation

BER validation : RIS multipath is correctly captured



- Baselines : theoretical Rayleigh Without Sionna and Flat Fading With Sionna overlap, as before.
- Sionna + RIS departs from both, requiring higher E_s/N_0 for the same BER, across every QAM order.
- Confirms the RIS's multipath is correctly captured, rather than collapsing back to the flat-fading case.

Thesis Vision Board (3 Years)

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- State-of-the-art review
- Problem formulation
- Q-Learning for RIS beamforming, 1st publication (WCNC 2025)
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- Journal submission
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Future Perspectives

■ *Mobile user scenario :*

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- Handle user mobility and time-varying channels.
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■ *Multi-user extension :* Scale the study to multi-user scenarios.

Scientific contributions



T. Chêne, O. Bounhar, G. R.-B. Othman, and O. Damen, “Adaptive Passive Beamforming in RIS-Aided Communications with Q-Learning,” in **2025 IEEE Wireless Communications and Networking Conference (WCNC)**. Milan, Italy : IEEE, 2025, pp. 1–6.

Thank you ! Questions ?

Oumaïma Bounhar

Supervised by Pr. **Ghaya Rekaya-Ben Othman**

Télécom Paris, Institut-Mines-Télécom, Palaiseau, France



22 septembre 2026

Doctoral school training requirements

Transversal skills

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Goal : minimize the number of pilots L_p (pilot overhead) while reliably finding Φ_{opt} .



System Model

Codebook design

We use a DFT codebook, and we separate the codebook into two parts :



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What is the configuration of the RIS that maximises the rate ?

State-of-the-art

Bayesian Classifier : Protocol

For a channel realization $\mathbf{H}$, we define $\Phi(\mathbf{H})$ as the class that assigns for this channel the best communication codeword $\Phi \in \Phi_1^{N_c}$:

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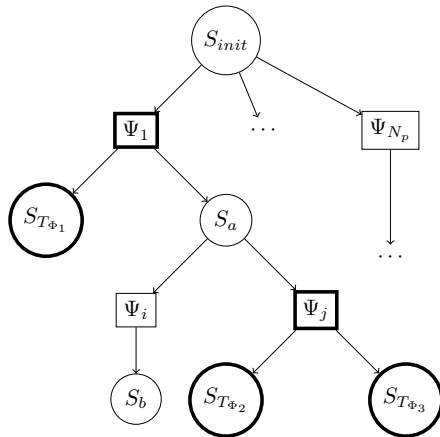
$$Y_k = \mathbf{y}_k \mathbf{y}_k^H$$

At each time step t , update the posteriori probability with knowledge over all received feedbacks :

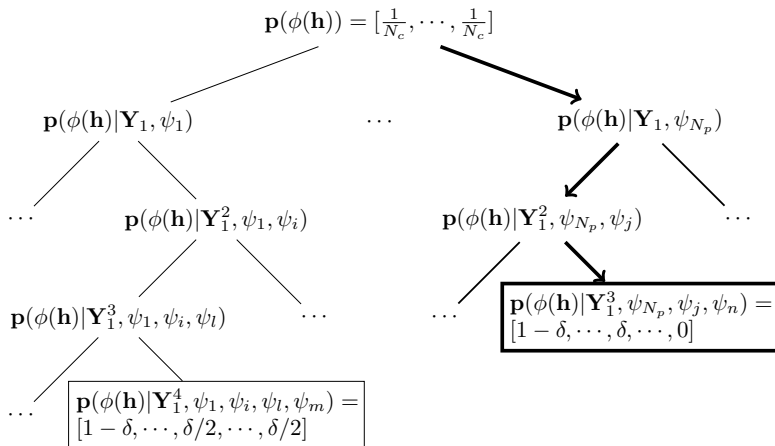
$$\mathbf{p}(\Phi(\mathbf{H})|Y_1^t, \Psi_1^t) \triangleq [\mathbf{p}(\Phi_1|Y_1^t, \Psi_1^t), \dots, \mathbf{p}(\Phi_{N_c}|Y_1^t, \Psi_1^t)]$$

Proposed approach : Reinforcement Learning

Stochastic shortest path problem



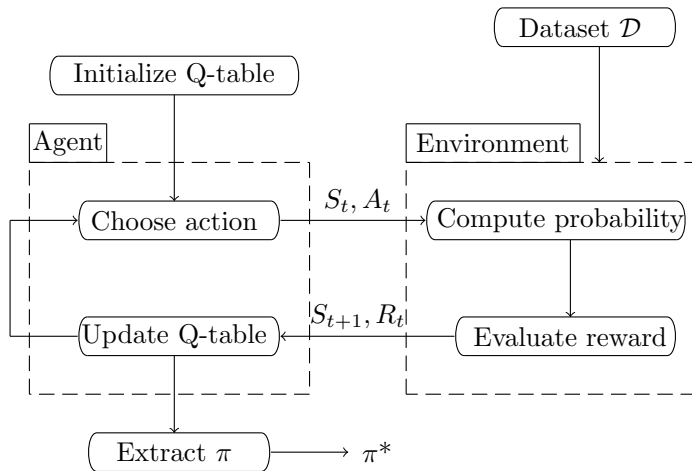
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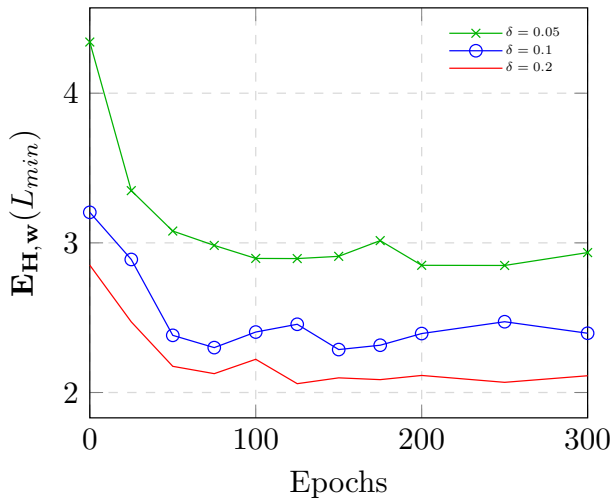
Euclidean distance

$$\mathbf{p}(\Phi(\mathbf{H}) = \Phi_k | \mathbf{Y}_1^{L_p}, \Psi_1^{L_p}) \propto \mathbf{p}(\Phi(\mathbf{H}) = \Phi_k | \mathbf{Y}_1^{L_p-1}, \Psi_1^{L_p-1}) \cdot \sum_{\mathbf{H}_1 / \mathbf{S}_T(\mathbf{H}_1)[k]=1} \frac{1}{(\mathbf{X}(\mathbf{H}_1)[L_p] - \mathbf{Y}_{L_p})^2} \quad (1)$$

Q-Learning algorithm



Stochastic shortest path problem





State-of-the-art

Blind Beamforming

Search for the optimal configuration (codeword $\Phi_{\text{opt}} \in \Phi_1^{N_c}$) of RIS over the possible configurations (communication codebook $\Phi_1^{N_c}$) :

- Exhaustive search over the codebook.
 - Huge overhead



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- Exhaustive search over the codebook.
 - Huge overhead
- Hierarchical : Need to create a specific codebook.
 - Low overhead.
 - Might not reach the optimal configuration because of its shape (binary tree).

State-of-the-art

Naive Bayes Classifier

Classification

Let L_p be the amount of the received samples, the Bayes optimal classifier is :

$$\operatorname{argmax}_{c \in \llbracket 1, N_c \rrbracket} \mathbf{p}(\Phi(\mathbf{H}) | Y_1^{L_p}, \Psi_1^{L_p})$$



State-of-the-art Naive Bayes Classifier

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Class c_i is the class assigned to a UE direction for which communication codeword Φ_i maximizes the received signal strength $\|\mathbf{h}(\Phi_i)\|_2^2$.

State-of-the-art

Naive Bayes Classifier

Objective

Our main purpose is to find the optimal configuration Φ_{opt} while minimizing the number of required pilots $\Psi \in \Psi_1^{N_p}$:

$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} = 1\}$$

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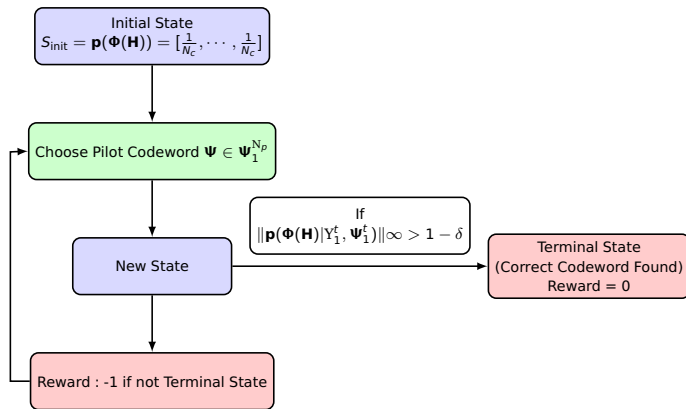
$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} = 1\}$$

We introduce a fixed degree of precision $\delta \in [0; 1]$ and we need to minimize L_p st. the probability of correct classification respects a certain precision :

$$L_{\min} = \min\{L_p : \|\mathbf{p}(\Phi(\mathbf{H})|Y_1^{L_p}, \Psi_1^{L_p})\|_{\infty} > 1 - \delta\}$$

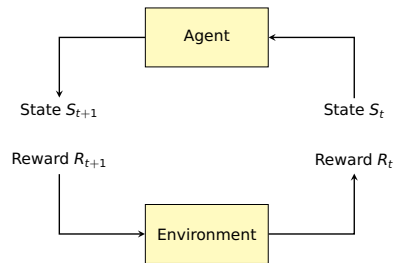
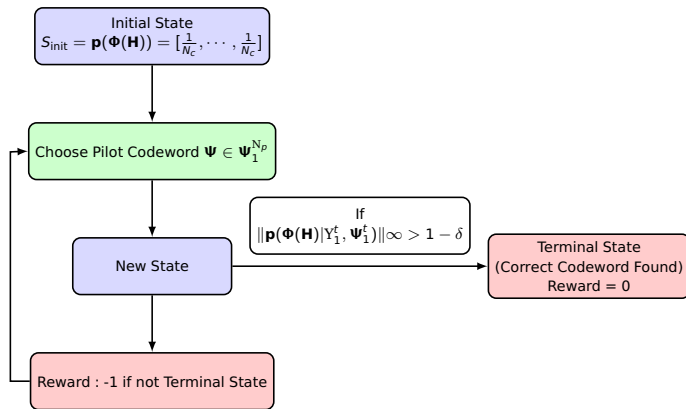
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State, Action, and Reward Flow



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A policy $\pi(S)$ determines which action A is chosen in a given state S :

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State-of-the-art

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$$\pi(S) = A$$

Objective

The goal is to reach a terminal state where the probability of correct classification respects a fixed degree of precision $\delta \in [0; 1]$:

$$L_{\min} = - \sum_{t=0}^{\infty} R_t$$

State-of-the-art

Q-Learning : Value Function and Policy

Q-function

The Q-value function represents the expected cumulative reward when taking action A_t in state S_t :

$$Q(S_t, A_t) = \mathbb{E}_{\mathbf{H}, \mathbf{w}} \left[\sum_{k=0}^{\infty} R_{t+k+1} \mid S_t = S, A_t = A \right]$$

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Bellman Equation

The Q-values are updated using the Bellman equation :

$$Q(S_t, A_t) \leftarrow (1 - \alpha)Q(S_t, A_t) + \alpha[R_{t+1} + \max_A Q(S_{t+1}, A)]$$

where $\alpha \in [0, 1]$ is the learning rate.

Optimal Policy

The optimal policy is obtained by selecting the action that maximizes the expected reward :

$$\pi^*(S) = \arg \max_A Q^*(S, A)$$

State-of-the-art

Q-Learning : Optimal policy

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Objective

Thus, our goal is to find the optimal policy π^* that maximizes the cumulative reward despite randomness in the channel and noise. We rewrite our objective :

$$L^* = \max_{\pi} \mathbb{E}_{\mathbf{H}, \mathbf{w}} \left[\sum_{t=0}^{\infty} R_t \right]$$



Proposed approach : Reinforcement Learning

From Q-Learning to Deep Q-Learning

- The state space was discretized [1];

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From Q-Learning to Deep Q-Learning

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- Transitions were determined using Euclidean distances to predefined states ;
- This approach became unreliable in high dimensions ;
- Small variations led to misclassifications among similar codewords.

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Deep Q-Neural Network

Deep Q-Learning (DQL) uses two NN :

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- Evaluation network (Eval-Net) parameterized by the NN weights $\theta^{(eval)}$,
- Target network (Target-Net) parameterized by NN weights $\theta^{(target)}$.

Approximation of the Q-function

The Q-function approximation can thus be expressed as :

$$Q_{eval}(S, A; \theta^{(eval)}) \approx \mathbb{E} \left[R + \gamma \max_{A'} Q_{target}(S', A'; \theta^{(target)}) \right], \quad (2)$$

Proposed approach : Reinforcement Learning

Deep Q-Learning Algorithm

Epsilon-Greedy Exploration: Agent selects action: $\arg \max Q_{eval}$ with prob. $(1 - \epsilon)$, random with prob. ϵ

Experience Replay: Store transitions (S, A, R, S') in buffer $\mathcal{M}$ and sample mini-batches

Target Network Update: Periodically copy $\theta^{eval} \rightarrow \theta^{target}$ for stable learning

Proposed approach : Reinforcement Learning

Deep Q-Neural Network

Network Optimization

The Q-network is trained by minimizing the mean squared error (MSE) loss, defined as :

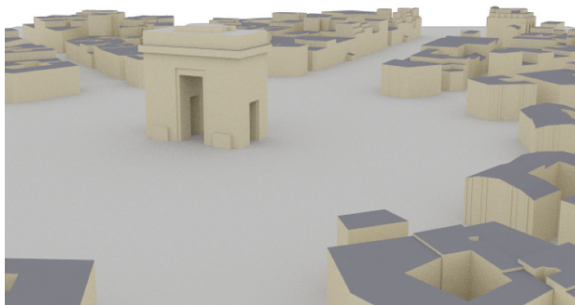
$$\mathcal{L}(\theta) = \mathbb{E}[(Q_{\text{target}} - Q_{\text{eval}}(S, A; \theta^{(\text{eval})}))^2], \quad (3)$$

with :

$$Q_{\text{target}} = R + \gamma \max_{\psi} Q_{\text{target}}(S', \psi; \theta^{(\text{target})}). \quad (4)$$

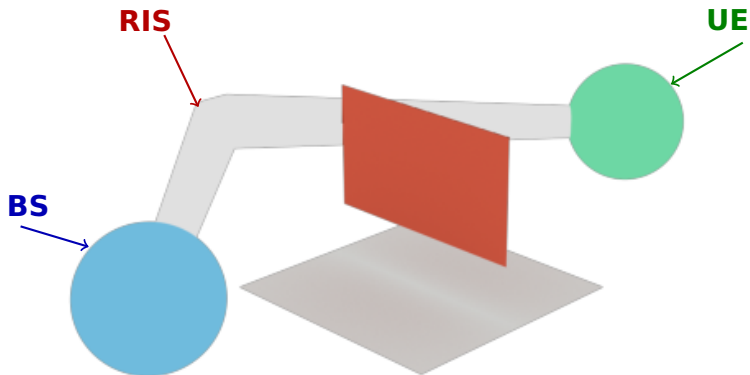
Sionna - Scene view

The area around the Arc de Triomphe in Paris



Sionna - Scene view

Simple floor-wall scenario



Visualization of the simulated scenario with a BS (64 antennas),
a RIS (1D 100 elements), and a UE (a single antenna).

Sionna - Realistic channel generation and validation

Primer on electromagnetics : RIS reradiated field

The reradiated field from the RIS at point S' is given by :

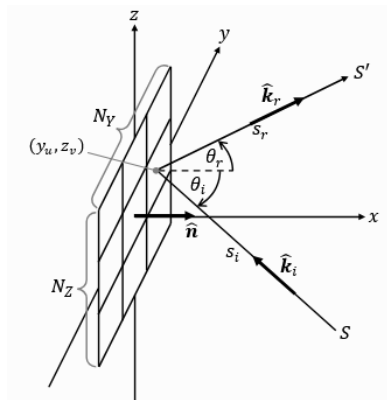
$$\mathbf{E}_r(S') = \sum_{u=1}^{N_y} \sum_{v=1}^{N_z} \Gamma_{u,v} G_{u,v} A_{u,v} \mathbf{p}_{u,v}$$

where :

$$G_{u,v} = \frac{3\lambda}{16\pi} (1 + \cos \theta_i^{u,v}) (1 + \cos \theta_r^{u,v}) ,$$

$$A_{u,v} = \frac{e^{-jk_0(s_i^{u,v} + s_r^{u,v})}}{s_i^{u,v} s_r^{u,v}} ,$$

$$\mathbf{p}_{u,v} = E_{i,\theta}(S) \hat{\theta}(\hat{\mathbf{k}}_r^{u,v}) + E_{i,\varphi}(S) \hat{\varphi}(\hat{\mathbf{k}}_r^{u,v})$$



Incident and reradiated field from a RIS

$$s_i^{u,v} = \|\mathbf{r}_{\text{RIS}}^{u,v} - \mathbf{r}_{\text{Tx}}\|_2$$

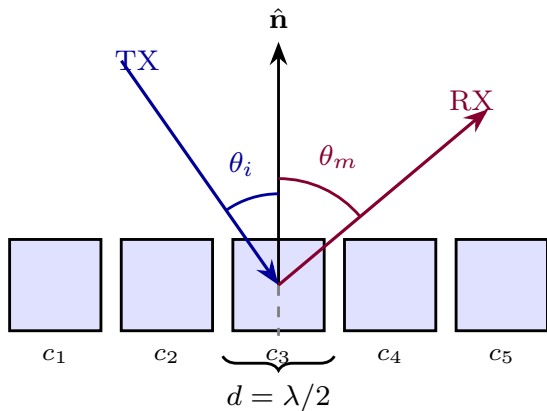
Tx-RIS distance

$$s_r^{u,v} = \|\mathbf{r}_{\text{Rx}} - \mathbf{r}_{\text{RIS}}^{u,v}\|_2$$

RIS-Rx distance

Sionna - Realistic channel generation and validation

Unit-cell geometry : incidence and reradiation angles



- Cells are spaced by half a wavelength, $d = \lambda/2$.
- Each cell sees its **own** local incidence angle θ_i and reradiation angle θ_m w.r.t. the surface normal $\hat{n}$.
- The per-cell gain scales with $(1 + \cos \theta_i)(1 + \cos \theta_m)$, and the path loss with $1/(s_i s_m)$, the Tx-cell and cell-Rx distances.

Sionna - Realistic channel generation and validation

Why geometry strongly affects the RIS channel

$$\text{Path loss} \propto \frac{1}{s_i^{u,v} s_r^{u,v}}$$

Main message

For RIS-assisted propagation, the attenuation is controlled by the both segments Tx-RIS and RIS-Rx, and it is proportional to the product of these two distances :

$$d_{\text{Tx} \rightarrow \text{RIS}} \times d_{\text{RIS} \rightarrow \text{Rx}}.$$



Sionna - Realistic channel generation and validation

Key takeaways

- In far-field regimes, per-cell delay/angle differences are tiny $\Rightarrow$ behaves like array/beamforming gain.



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Key takeaways

- In far-field regimes, per-cell delay/angle differences are tiny $\Rightarrow$ behaves like array/beamforming gain.
- In near-field regimes (large RIS, close TX/RX), delay and angular spread become physically significant $\Rightarrow$ **resolvable** multipath.